「Consider the curve along the unit circle,

$$z(t) = e^{it} \quad (0 \le t \le 2\pi).$$

By definition of trigonometric,

$$\sin t = \frac{e^{it} - e^{-it}}{2i} = \frac{1}{2i}\cdot\left(z - \frac{1}{z}\right) \text{ and } \cos t = \frac{e^{it} + e^{-it}}{2} = \frac{1}{2}\left(z + \frac{1}{z}\right)$$

Using this facts, evaluate the integral,

$$\int_0^{2\pi} g(\sin\theta, \cos\theta)\, d\theta \text{ where } g \text{ is a rational function.}$$

「Evaluate : for real numbers a, b with $|a| > |b| > 0$.

$$I = \int_0^{2\pi} \frac{1}{a + b\cdot\cos\theta}\, d\theta = \frac{2\pi}{\sqrt{a^2 - b^2}}.$$

「since $|a| > |b|$, $I = \frac{1}{b}\cdot\int_0^{2\pi} \frac{1}{a/b + \cos\theta}\, d\theta = \frac{1}{b}\cdot\int_0^{2\pi} \frac{1}{\alpha + \cos\theta}\, d\theta$ where $\alpha = \frac{a}{b}$, $|\alpha| > 1$.

By the fact above, along $C : z(\theta) = e^{i\theta}$ $(0 \le \theta \le 2\pi)$,

$$\cos\theta = \frac{1}{2}\left(z + \frac{1}{z}\right) \text{ and } dz = i\cdot e^{i\theta} d\theta \Rightarrow d\theta = \frac{1}{ie^{i\theta}} dz = \frac{1}{i\cdot z} dz.$$

Therefore, $\displaystyle\int_0^{2\pi} \frac{1}{\alpha + \cos\theta}\, d\theta = \int_C \frac{1}{\alpha + \frac{1}{2}\cdot\left(z + \frac{1}{z}\right)}\frac{1}{iz}\, dz = \frac{2}{i}\cdot\int_C \frac{1}{z^2 + 2\alpha z + 1}\, dz$

Let $f(z) = \frac{1}{z^2 + 2\alpha z + 1}$. Since $z^2 + 2\alpha z + 1$ has two roots, $-\alpha \pm \sqrt{\alpha^2 - 1}$,

If $\alpha > 1$, then $-\alpha + \sqrt{\alpha^2 - 1}$ lies inside C, but $-\alpha - \sqrt{\alpha^2 - 1}$ lies outside of C.

$$(-\alpha + \sqrt{\alpha^2 - 1} < -\alpha + \sqrt{\alpha^2} = 0, \text{ but } (-1 < -\alpha + \sqrt{\alpha^2 - 1}).$$

If $\alpha < -1$, similarly $-\alpha + \sqrt{\alpha^2 - 1}$ lies outside C, $-\alpha - \sqrt{\alpha^2 - 1}$ lies inside of C.

So in case of $\alpha > 1$, $f(z)$ has only one isolated singularity $-\alpha + \sqrt{\alpha^2 - 1}$ as a simple pole.

therefore the Residue Theorem gives that

$$\int_0^{2\pi} \frac{1}{\alpha + \cos\theta}\, d\theta = \frac{2}{i}\cdot\int_C f(z)\, dz = \frac{2}{i}\cdot 2\pi i\cdot \lim_{z \to -\alpha + \sqrt{\alpha^2 - 1}} \left(z - (-\alpha + \sqrt{\alpha^2 - 1})\right)\cdot\frac{1}{z^2 + 2\alpha z + 1}$$

$$= 4\pi\cdot\frac{1}{2\sqrt{\alpha^2 - 1}} = \frac{2\pi}{\sqrt{\alpha^2 - 1}}.$$ Consequently

$$\int_0^{2\pi} \frac{1}{a + b\cos\theta}\, d\theta = \frac{1}{|b|}\cdot\frac{2\pi}{\sqrt{(a/b)^2 - 1}} = \frac{2\pi}{\sqrt{a^2 - b^2}}.$$

Similarly, we can deduce that in case of $\alpha < -1$, as $= \frac{-2\pi}{\sqrt{a^2 - b^2}}$.

[Subsequentially, we can deduce that

$$I = \int_0^{\pi} \frac{1}{a^2 + \cos^2\theta}\, d\theta = \int_0^{\pi} \frac{1}{a^2 + \sin^2\theta}\, d\theta = \frac{\pi}{a\sqrt{1+a^2}} \quad \text{for } a>0.$$

Using the useful formulars;

$$2\cdot\cos^2\theta = 1 + \cos 2\theta \quad \text{and} \quad 2\cdot\sin^2\theta = 1 - \cos 2\theta.$$

(Note: it can be deduced the $\cos(\alpha+\beta) = \cos\alpha\cdot\cos\beta - \sin\alpha\sin\beta$.)

Now,
$$I = \int_0^{\pi} \frac{1}{a^2+\cos^2\theta}\, d\theta = \int_0^{\pi} \frac{2}{1+2a^2+\cos 2\theta}\, d\theta \qquad \text{so, } 1+2a^2>1,$$

$$= \int_0^{2\pi} \frac{1}{1+2a^2+\cos u}\, du = \frac{2\pi}{\sqrt{(1+2a^2)^2-1}} = 2\pi\cdot\frac{1}{\sqrt{4a^4+4a^2}}$$

$$= \frac{\pi}{a\sqrt{1+a^2}}.$$

「Evaluate

$$\int_0^{2\pi} \frac{1}{1-2\alpha\cdot\cos\theta+\alpha^2}\,d\theta = \frac{2\pi}{|\alpha^2-1|} \quad (\alpha \neq \pm 1,\ \text{real.}).$$

Proof. Notice that: for $z(\theta) = e^{i\theta}$, $\cos\theta = \dfrac{e^{i\theta}+e^{-i\theta}}{2} = \dfrac{1}{2}\cdot\left(z+\dfrac{1}{z}\right)$.

And, $dz = i\cdot e^{i\theta}d\theta \Rightarrow d\theta = -i\cdot z^{-1}$, so

$$\int_0^{2\pi} \frac{1}{1-2\alpha\cdot\cos\theta+\alpha^2}\,d\theta = \int_C \frac{-i}{1-\alpha\cdot(z+z^{-1})+\alpha^2}\cdot\frac{1}{z}\,dz$$

$$= i\cdot\int_C \frac{1}{\alpha\cdot z^2-(1+\alpha^2)\cdot z+\alpha}\,dz$$

$$= \frac{i}{\alpha}\cdot\int_C \frac{1}{z^2-(\alpha+\frac{1}{\alpha})z+1}\,dz$$

Since $z^2-(\alpha+\frac{1}{\alpha})z+1 = (z-\alpha)(z-\frac{1}{\alpha})$, consider two case:

· $-1<\alpha<1$ or $\alpha>1$ or $\alpha<-1$. (if $\alpha=0$, then it is just constant integral).

If $-1<\alpha<1$, then the function in the integral has a simple pole at α inside C, So $\operatorname{Res}(f,\alpha) = \lim_{z\to\alpha}(z-\alpha)\cdot\dfrac{1}{(z-\alpha)(z-\frac{1}{\alpha})} = \dfrac{1}{\alpha-\frac{1}{\alpha}} = \dfrac{\alpha}{\alpha^2-1}$.

Now, by the Residue theorem,

$$\int_0^{2\pi} \frac{1}{1-2\alpha\cos\theta+\alpha^2}\,d\theta = \frac{i}{\alpha}\cdot\int_C \frac{1}{z^2-(\alpha+\frac{1}{\alpha})z+1}\,dz = \frac{i}{\alpha}\cdot 2\pi i\cdot\frac{\alpha}{\alpha^2-1} = \frac{2\pi}{1-\alpha^2}.$$

If $\alpha>1$, then the function has a simple pole at $\frac{1}{\alpha}$, inside C, so

$$\operatorname{Res}(f,\tfrac{1}{\alpha}) = \lim_{z\to\frac{1}{\alpha}}(z-\tfrac{1}{\alpha})\cdot\frac{1}{(z-\alpha)(z-\frac{1}{\alpha})} = \frac{1}{\frac{1}{\alpha}-\alpha} = \frac{\alpha}{1-\alpha^2}.$$

therefore, by the Residue theorem again,

$$\text{〃} = \frac{2\pi}{\alpha^2-1}.$$

If $\alpha<-1$, we obtain the value, similarly $\alpha>1$ case.

Consequently, $\displaystyle\int_0^{2\pi} \frac{1}{1-2\alpha\cdot\cos\theta+\alpha^2}\,d\theta = \frac{2\pi}{|\alpha^2-1|}$ ☒

「Evaluate

$$\int_0^{\frac{\pi}{2}} \frac{1}{1+\sin^2\theta}\, d\theta = \frac{\pi}{2\sqrt{2}}$$

Proof.